Ideation Phase

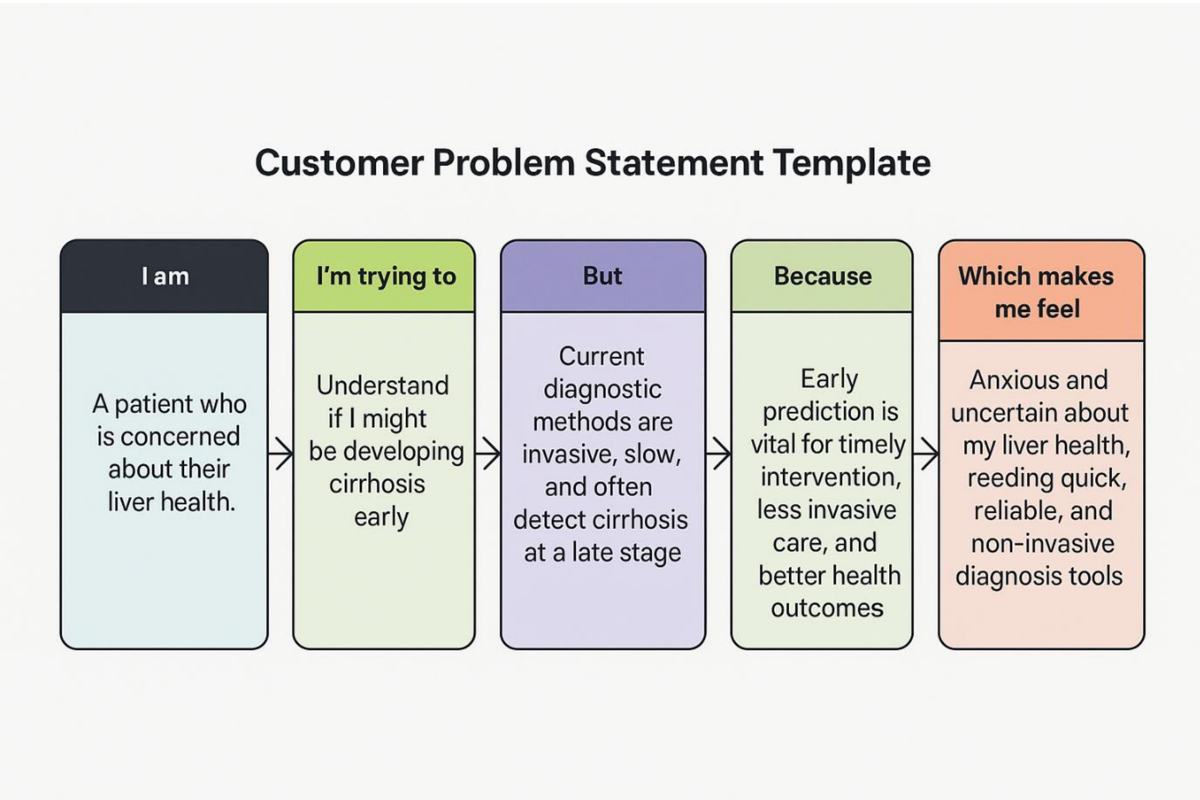
Define the Problem Statements

Date	14 feb 2026
Team ID	LTVIP2026TMIDS35912
Project Name	Deep Learning Fundus Image Analysis for Early Detection of Diabetic Retinopathy
Maximum Marks	2 Marks

Customer Problem Statement Template:

Diabetic retinopathy is a serious eye disease caused by prolonged diabetes and often remains undetected until it reaches an advanced stage, leading to irreversible vision loss. Early diagnosis is crucial for effective treatment and prevention of blindness; however, traditional screening methods are time-consuming, costly, and require specialized ophthalmologists and equipment.

This project aims to improve eye care by applying advanced deep learning techniques to predict diabetic retinopathy from retinal fundus images in a non-invasive manner. By identifying hidden patterns and abnormalities in retinal images, the system delivers accurate early-stage predictions, enabling timely medical intervention and supporting personalized, accessible, and cost-effective healthcare solutions.



PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A diabetic patient concerned about eye health	Understand if I am developing diabetic retinopathy at an early stage	Early stages show no visible symptoms and screening requires specialist examinati on	Early detection is critical to prevent vision loss and blindness	Anxious and uncertain about my eye health, needing a quick and reliable screening method
PS-2	A long-term diabetic with a family history of eye disease	Know my risk of developing diabetic retinopathy early	Regular eye screenings are costly and not easily accessible	Knowing the risk early helps in lifestyle control and timely treatment	Worried and helpless due to lack of affordable and accessible eye screening
PS-3	A healthcare provider treating diabetic patients	Detect early-stage diabetic retinopathy efficiently	Manual fundus image analysis is time-consuming and prone to human error	Automated, accurate predictions improve diagnosis and patient care	Frustrated and overburdened due to increasing patient load and delayed diagnosis
PS-4	A diabetic patient experiencing blurred vision or eye strain	Understand whether symptoms indicate diabetic retinopathy	Symptoms are vague and confirmati on requires specialist evaluation	A data-driven image analysis system can provide fast and accurate insights	Anxious and confused, needing clarity without delay or invasive procedures